SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 5

Cost-Volume-Profit, Variable Costing & Activity-Based Costing

Managerial accounting answers three questions: what happens when volume changes (CVP)? which costs are product costs (variable vs absorption)? which products/customers actually consume resources (ABC)? Includes Java Roasters, Classic Brass, Meridian and Blue Harbor exercises.

Material

1 CVP: five moving parts and contribution margin

CVP links price, variable cost, fixed cost, volume and sales mix. `CM = Sales − Variable costs` covers fixed costs first, then becomes profit. Racing Bicycle: price $500, VC $300, unit CM $200, **CM ratio 40%**, fixed $80k; each extra $1 of sales adds $0.40 CM.

2 Break-even, target profit, margin of safety, operating leverage

Break-even = Fixed ÷ unit CM = 400 bikes = Fixed ÷ CM ratio = $200k — a floor, not a goal. Target profit units = (Target + Fixed) ÷ unit CM = 900 bikes.

Margin of safety = actual − break-even sales (e.g. $12,500 = 12.5%).

DOL = CM ÷ operating income; %ΔNOI = DOL × %ΔSales. High fixed costs (DOL 4.0): profit $8k–$72k on ±20% sales; low fixed (DOL 2.5): $20k–$60k. Leverage cuts both ways.

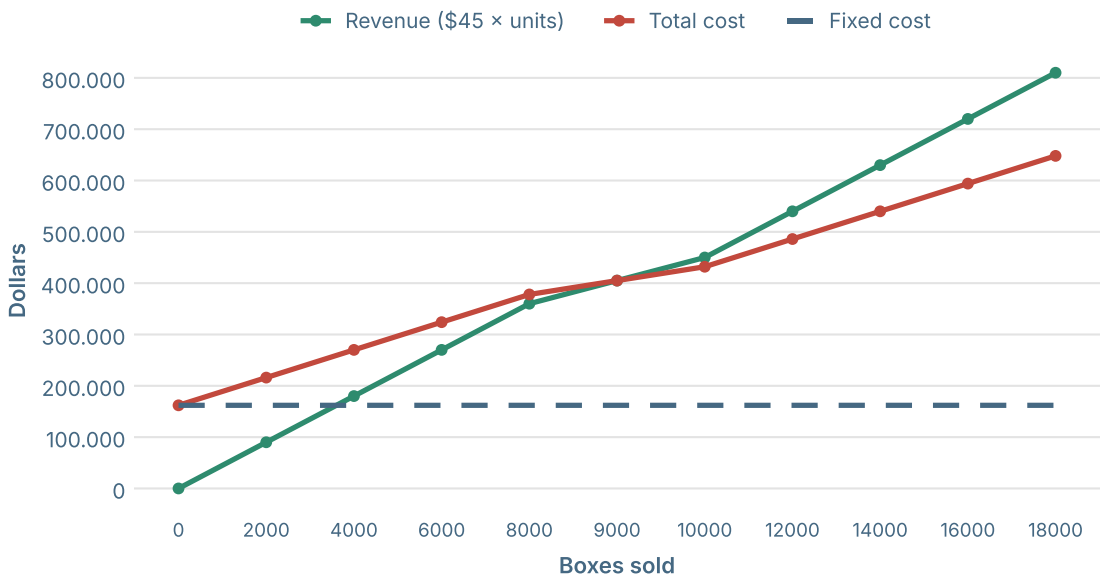
Sales mix: weighted CM per bundle ($70) → 5,000 bundles = 10,000 A + 15,000 B; a shift to low-CM products raises break-even.

Assumptions: constant price, linear costs, constant mix, no inventory change. Use CVP as a model, not a guarantee.

3 See it: the CVP graph and a break-even calculator Deep dive

Revenue starts at zero with slope = price; total cost starts at fixed cost with slope = variable cost. Where they cross is break-even.

Java Roasters CVP graph



Lines meet at 9,000 boxes ($405,000); at 12,000 boxes the gap is the $54,000 profit.

Try it with Java Roasters' numbers ($45 price, $27 variable cost, $162,000 fixed cost).

Reading DOL

DOL = total contribution ÷ profit; it rises sharply near break-even, so profit becomes very sensitive.

4 Variable vs absorption costing, and segment reporting

Listed in this week's map; summarized from standard managerial accounting.

The only difference is **fixed manufacturing overhead**: absorption costing inventories it (external reporting); variable costing expenses it (internal decisions, consistent with CVP). When production exceeds sales, absorption profit is higher — a lever that can be abused.

Segment reporting: segment margin = segment CM − traceable fixed costs; common costs remain if a segment closes, so judge segments on segment margin.

5 Why variable and absorption costing profits differ

Deep dive

The only difference is fixed manufacturing overhead (FMOH): absorption costing stores part of it in inventory when production exceeds sales; variable costing expenses it all.

Condition	Higher profit
Production > sales	Absorption
Production < sales	Variable
Equal	Same

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Profit difference = FMOH per unit × change in inventory units
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Example

FMOH $10/unit, inventory up 4,000 units → absorption profit $40,000 higher.

Example: absorption vs variable profit as production changes

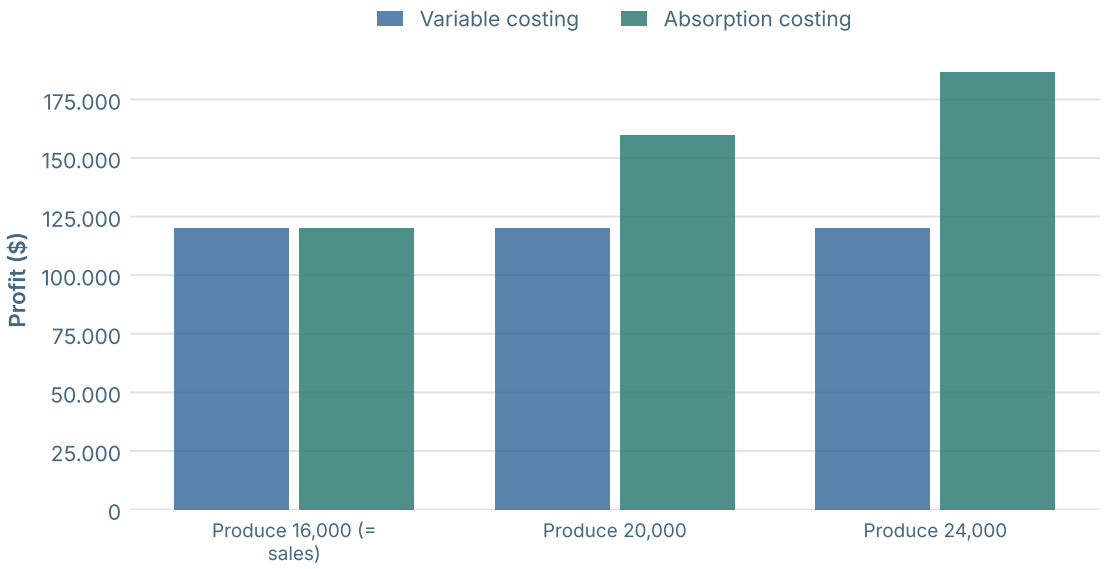


Illustration: sales fixed at 16,000 units, FMOH $200,000. More production parks more FMOH in inventory.

Watch out

Absorption profit can be "boosted" by overproducing.

6 Activity-Based Costing (ABC)

Traditional systems use one rate; ABC asks what activities cause cost: resources → activities → cost objects. ABC assigns some nonmanufacturing costs, excludes organization-sustaining costs, and uses many pools. Hierarchy: unit, batch, product, customer, organization-sustaining. Five steps: define activities, assign costs to pools, compute rates, assign to objects, report. Limitations: cost, resistance, misinterpretation, non-GAAP, subjective allocations — always ask whether the cost will actually change.

7 ABC in one picture: where overhead really comes from Deep dive

Resources → Activities → Activity drivers → Products / customers

Traditional costing uses one driver (machine hours) for all overhead; custom products that use many orders and designs are undercosted.

Classic Brass: product margin, traditional vs ABC



The 'profitable' custom product is actually loss-making once design and order costs are traced.

+$290,500
Standard Stanchions margin after ABC

−$307,500
Custom Compass Housings margin (now a $49,500 loss)

$1,470
Customer relations cost per customer

Cost hierarchy



Key takeaways

1. Contribution margin covers fixed costs, then becomes profit.
2. Break-even = Fixed ÷ unit CM (or ÷ CM ratio); target units = (Target + Fixed) ÷ unit CM.
3. Margin of safety = sales cushion before loss.
4. DOL = CM ÷ operating income; high fixed costs amplify swings.
5. A shift to low-CM products raises break-even.
6. Absorption vs variable costing differ only in fixed overhead.
7. ABC: resources → activities → cost objects; often reveals loss-making custom products/customers.

Case studies

Case & worked answers

CVP exercises: Java Roasters

Unit CM $18; CM ratio 40%; break-even 9,000 boxes = $405,000; target 14,000 boxes. At 12,000 boxes: MOS 3,000 = $135,000 = 25%; NOI $54,000; DOL 4.0; +15% sales → +60% NOI = $86,400 (verified).

Case & worked answers

Classic Brass: ABC reverses product profitability

Activity rates: $320/order, $630/design, $19/MH, $1,470/customer. ABC overhead: Standard $524,500; Custom $427,500. Margins: Standard $615,750 → **$906,250**; Custom $258,000 → **($49,500)**. Traditional MH-based costing over-costed the high-volume product and under-costed the custom one, which consumes all design work. Windward Yachts customer margin: $699.

Case & worked answers

Exercises 3 & 4: Meridian Furniture and Blue Harbor Fittings

Meridian: rates $150/order, $300/setup, $15/MH; ABC Chairs $540k, Desks $330k; traditional $29/MH → Chairs $696k, Desks $174k. Desks are undercosted — 75% of setups, 20% of machine hours.

Blue Harbor: activity cost $8,920; total $12,770; margin ($2,970). Many small orders plus 3 custom designs. Actions: minimum order size, separate design fees, lower service, reassess the relationship.

Worked examples

Problem 1 · Easy **Break-even dan target laba**

Harga Rp50, biaya variabel Rp30 per unit, biaya tetap Rp200.000. Hitung (a) BEP unit dan rupiah, (b) unit untuk laba Rp60.000.

1. Unit CM = 50 − 30 = 20; CM ratio = 20 ÷ 50 = 40%.
2. `BEP unit = 200.000 ÷ 20 = 10.000 unit; BEP Rp = 200.000 ÷ 0,40 = Rp500.000`
3. `Unit target = (60.000 + 200.000) ÷ 20 = 13.000 unit`

Answer: BEP 10.000 unit (Rp500.000); 13.000 unit untuk laba Rp60.000.

Problem 2 Medium **MOS dan DOL**

Lanjutan: penjualan aktual 12.000 unit. Hitung laba, margin of safety, DOL, dan laba baru jika penjualan naik 10%.

1. CM total = 12.000 × 20 = 240.000 → laba = 240.000 − 200.000 = 40.000.
2. MOS = 12.000 − 10.000 = 2.000 unit; MOS% = 2.000 ÷ 12.000 = 16,7%.
3. `DOL = CM ÷ Laba = 240.000 ÷ 40.000 = 6`
4. Penjualan +10% → laba +6 × 10% = +60% → 40.000 × 1,6 = 64.000. Cek: 13.200 × 20 − 200.000 = 64.000 ✓

Answer: Laba Rp40.000; MOS 16,7%; DOL 6; laba naik ke Rp64.000.

Problem 3 Challenging **Absorption vs variable costing**

Produksi 10.000 unit, terjual 8.000 unit, overhead tetap Rp100.000. Metode mana yang labanya lebih tinggi, dan selisihnya berapa?

1. Overhead tetap per unit (absorption) = 100.000 ÷ 10.000 = 10.
2. Persediaan naik 2.000 unit. Di absorption, 2.000 × 10 = 20.000 overhead tetap "parkir" di persediaan (belum jadi beban).
3. Di variable costing, seluruh 100.000 langsung jadi beban periode ini.

Answer: Absorption costing lebih tinggi Rp20.000 (= kenaikan persediaan × overhead tetap per unit).

Practice questions

Try answering first. The answer key is below each question.

1. Price $45, variable cost $27, fixed cost $162,000. Break-even units?

A. 3,600
B. 6,000
C. 9,000
D. 14,000

Answer: C. CM = 18; 162,000 ÷ 18 = 9,000.

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2. Sales 12,000, profit $54,000, contribution $216,000. If sales rise 10%, profit rises about...

A. 10%
B. 25%
C. 40%
D. 60%

Answer: C. DOL = 4, so 4 × 10% = 40%.

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3. Production exceeds sales. Which profit is higher?

A. Variable costing
B. Absorption costing
C. Equal
D. Cannot tell

Answer: B. Absorption parks some fixed overhead in inventory.

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4. In the ABC hierarchy, machine setup per batch is...

A. Unit-level
B. Batch-level
C. Product-level
D. Organisation-sustaining

Answer: B. Setup occurs once per batch regardless of units.

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5. Why do custom products often look profitable under traditional costing?

A. They always have higher prices
B. Overhead is spread by machine hours while custom products consume order and design activities
C. Lower material costs
D. High volumes

Answer: B. A single volume driver misses batch and design activities.

Sources

- P3MD Week 5 - CVP & ABC_AZ.pdf
- UPDATED -Week 5- MK02.pdf